

UDITYA NARAYAN TIWARI

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Uttar Pradesh, India

SUMMARY

AI/ML Engineer specializing in **GenAI**, **NLP**, and **end-to-end AI/ML applications**. Published two open-source Python libraries on **PyPI** with **16,500+** downloads worldwide. Experienced in developing **RAG** pipelines, **semantic search**, and scalable **REST APIs** using LangChain, FAISS, ChromaDB, and **FastAPI**, with proficiency in **Docker** and **AWS**. Strong foundation in **DSA** and a passion for building production-ready AI solutions.

EDUCATION

VIT Bhopal University
B.Tech - CSE (AI & ML) | CGPA: 9.00 / 10.00

Sehore, Madhya Pradesh
2023 – Present

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI & Machine Learning: Machine Learning, Deep Learning, **NLP**, **Feature Engineering**, Model Evaluation, Predictive Analytics

Generative AI: LLMs, **RAG**, **Embeddings**, Semantic Search, **Vector Databases (FAISS, ChromaDB)**

Frameworks & Libraries: LangChain, LangGraph, Scikit-learn, TensorFlow, PyTorch, Sentence Transformers, FastAPI, Flask

Data Science: Pandas, NumPy, Matplotlib, Plotly

Cloud & Infrastructure: Amazon Web Services (AWS), Docker

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Kaggle

PROJECTS

NeuroLens: Knowledge Retrieval Engine — Python, LangChain, RAG, LLMs, FAISS, ChromaDB

[GitHub](#)

- Built a **Retrieval-Augmented Generation (RAG)** platform supporting **context-aware question answering** across PDFs, documents, and web content through Large Language Models (LLMs).
- Implemented semantic retrieval with vector embeddings in FAISS and ChromaDB, delivering **fast, accurate, and scalable knowledge retrieval**.
- Designed **automated document ingestion, indexing, and retrieval workflows**, significantly improving information discovery through intelligent contextual responses.

Smart Medical Care for Rural Areas — Python, NLP, Flask, Sentence Transformers

[GitHub](#)

- Designed an intelligent medicine recommendation system delivering **symptom-based medicine recommendations** through Natural Language Processing (NLP).
- Leveraged **cosine similarity** and **Maximal Marginal Relevance (MMR)** to improve semantic matching accuracy between symptoms and medicines.
- Deployed scalable **data preprocessing pipelines** and RESTful services with Flask for real-time medicine recommendations.

MultiModel Analysis Package — Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly, PyPI

[Documentation](#)

- Published an **open-source ML benchmarking framework** on **PyPI**, surpassing **6,500+** global downloads while reducing **80%** of the manual effort required for **model training, evaluation, and comparison**.
- Architected a **fault-tolerant benchmarking engine** supporting **15+** classification and regression algorithms with **automated feature scaling, data splitting, and resilient exception handling**.
- Introduced an **automated reporting and visualization pipeline** producing **ROC-AUC, Accuracy, F1-score, and R²** metrics with seamless export to **JSON, Excel**, and publication-ready charts.

EXTRACURRICULAR ACTIVITIES

Microsoft Learn Student Chapter
Technical Team Member & Community Admin

VIT Bhopal University, Sehore, MP
04/2025 – 04/2026

- Planned and executed **technical workshops, coding competitions, and hackathons**, promoting hands-on learning in Microsoft technologies, cloud computing, and AI.
- Led **community engagement initiatives** by coordinating with student developers, organizing technical sessions, and fostering knowledge sharing across the campus community.

ACHIEVEMENTS & CERTIFICATIONS

- Open Source Contributions:** Published and maintain two open-source Python libraries on **PyPI**, collectively surpassing **16,500+** downloads, including **10,000+** downloads for **NLP Text Preprocessing** and **6,500+** downloads for **MultiModel Analysis**.
- Microsoft Azure Data Fundamentals:** Earned Microsoft certification covering cloud computing fundamentals, Azure services, security, governance, and business applications.
- Indian Institute of Remote Sensing (IIRS), ISRO:** Successfully completed certification in **AI/ML for Geodata Analysis**, focusing on machine learning applications in remote sensing and geospatial intelligence.